

1. Let X denote number of raindrops that land on a detector during a certain fixed period of time. Suppose that X has a Poisson distribution with $E(X) = 5$.

1a. Calculate $P(X > 3)$.

1b. Calculate $P(X > 3 \mid X > 1)$.

2. Suppose that X_1, X_2, X_3, X_4, X_5 are independent Poisson random variables, with $E(X_j) = 0.7$ for each j . Define $X = X_1 + X_2 + X_3 + X_4 + X_5$. Calculate the value of $P(2 \leq X \leq 6)$.

3. Consider the 50,000 students on a college campus, and suppose that their dietary restrictions are independent. Suppose that a randomly selected person has a chocolate allergy with probability $\frac{1}{15,000}$. Find the approximate probability that 4 or more students on the campus have a chocolate allergy.

4. Consider X as defined in Question 1. Now define Y as a Geometric random variable, independent from X , with $p_Y(y) = (4/5)^{y-1}(1/5)$ for integers $y \geq 1$, i.e., with $E(Y) = 5$. Find $P(Y > X)$.

5. Consider the random variable X defined in Question 2. Find $E((X)(X-1)(X-2))$.